Experimental Design & Scope of Inference

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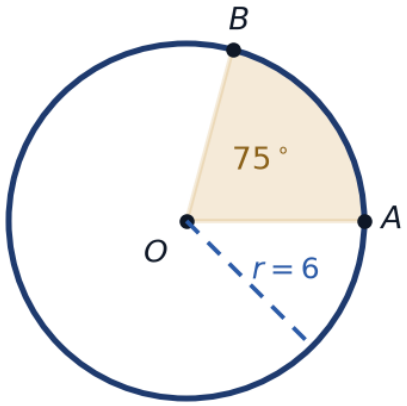
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Observational studies vs. experiments

A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.

2 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.



3 [Foundational] • Target: The three principles of experimental design (RRCB)

State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.

4 [Foundational] • Target: The three principles of experimental design (RRCB)

A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.

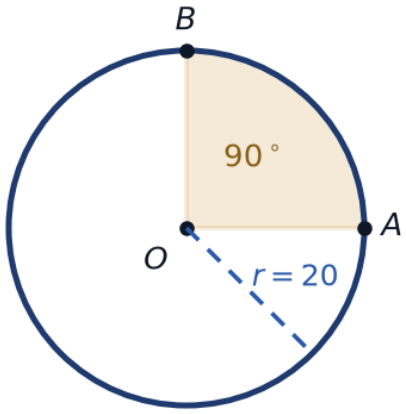
5 [Foundational] • Target: Blocking and matched pairs designs

A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.

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6 [Foundational] • Target: Blocking and matched pairs designs

A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.



7 [Foundational] • Target: Placebo effect and blinding

Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Placebo effect and blinding

Explain the difference between a single-blind and a double-blind experiment.

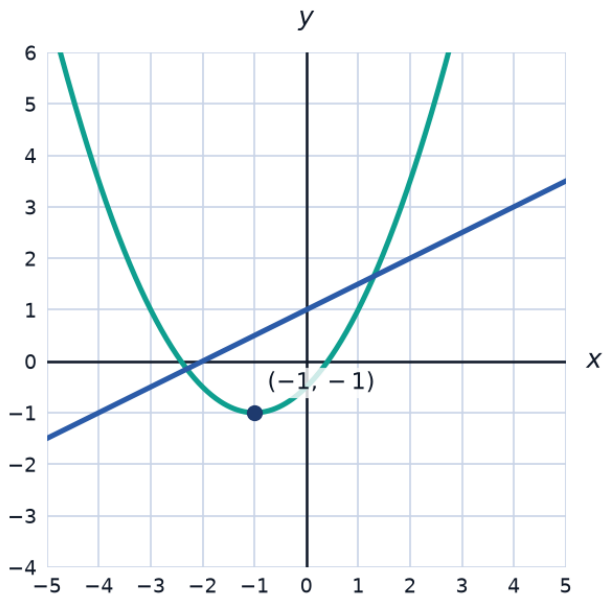
9 [Intermediate] • Target: Scope of inference

A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.

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10 [Intermediate] • Target: Scope of inference

A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.



11 [Intermediate] • Target: Putting it together: evaluating a study design

A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.

12 [Intermediate] • Target: Confounding variables

An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.

13 [Intermediate] • Target: Confounding variables

Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.

14 [Intermediate] • Target: Identifying experimental units and treatments

A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

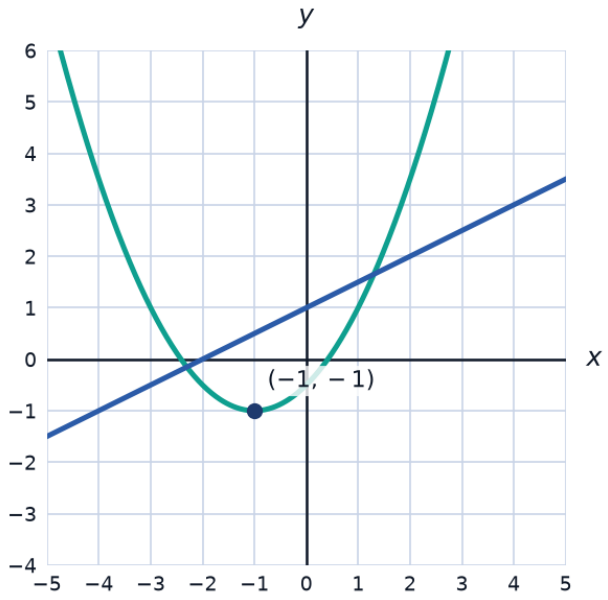
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15 **[Challenging]** • **Target: Identifying experimental units and treatments** • **Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.

16 **[Challenging]** • **Target: Identifying experimental units and treatments** • **Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.



17 **[Challenging]** • **Target: Identifying experimental units and treatments** • **Cross-Lesson: 1.01 Measures of Center & Spread**

A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.

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18 [Challenging] • Target: Putting it together: evaluating a study design • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A pharmaceutical company is investigating the efficacy of a new drug to reduce blood pressure. They want to design a randomized controlled experiment. Participants will be recruited from a specific patient population. To ensure generalizability and account for potential confounding factors, the researchers decide to stratify the participants by age group (Under 40, 40-60, Over 60) and by the severity of their hypertension (Mild, Moderate, Severe). If there are 150 participants in total and an equal number of participants are to be assigned to each treatment group (Drug A, Drug B, Placebo), and the number of participants in each stratum is proportional to the expected prevalence in the general population, how many participants would be allocated to the stratum 'Under 40' and 'Mild' hypertension if the proportion of individuals in this stratum is 0.08?

19 [Challenging] • Target: Confounding variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

20 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 2.01 Sampling Methods & Bias

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 467 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. Experiment

d. Meta-analysis

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22 [Foundational] • Target: The three principles of experimental design (RRCB)

A researcher randomly assigns 208 volunteers to receive a new exercise program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

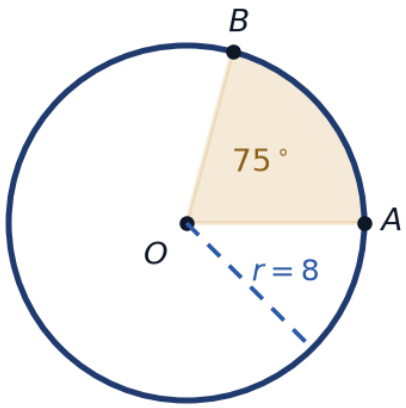
b. Census

c. Experiment

d. Meta-analysis

23 [Foundational] • Target: Blocking and matched pairs designs

A researcher randomly assigns 259 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.



a. Census

b. Meta-analysis

c. Observational study

d. Experiment

24 [Foundational] • Target: Placebo effect and blinding

A researcher randomly assigns 108 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Experiment

b. Census

c. Meta-analysis

d. Observational study

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25 [Foundational] • Target: Scope of inference

A researcher records data on exercise habits and cholesterol levels from 164 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study

b. Meta-analysis

c. Census

d. Experiment

26 [Foundational] • Target: Putting it together: evaluating a study design

A researcher records data on medication use and symptom severity from 391 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study

b. Experiment

c. Census

d. Meta-analysis

27 [Foundational] • Target: Confounding variables

A researcher records data on meditation habits and stress levels from 301 people without intervening in their lifestyle in any way. Identify the type of study.

a. Experiment

b. Observational study

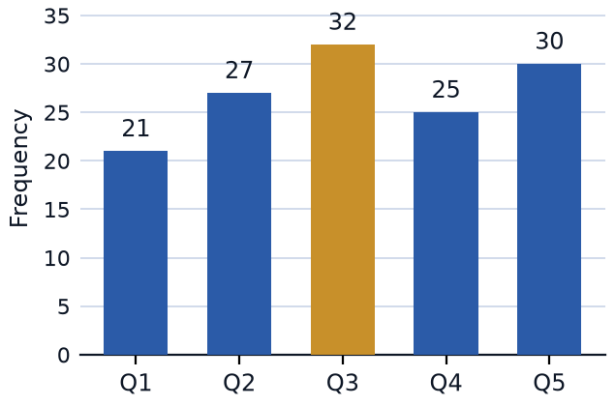
c. Census

d. Meta-analysis

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28 [Foundational] • Target: Identifying experimental units and treatments

In a study on medication use and symptom severity, a simple random sample of 130 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?



a. Causation, but only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population

c. Association only, and only about the subjects actually studied (not generalizable)

d. Association only, generalizable to the entire population

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Observational Studies vs. Experiments

In a study on tutoring attendance and exam performance, 127 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Association only, generalizable to the entire population

b. Causation, but only about the subjects actually studied (not generalizable)

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

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30 **[Intermediate]** • **Target: Principles of Experimental Design**

In a study on tutoring attendance and exam performance, 459 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

31 **[Intermediate]** • **Target: Blinding, Placebos, and Confounding**

In a study on tutoring attendance and exam performance, 149 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable)

b. Causation, but only about the subjects actually studied (not generalizable)

c. Association only, generalizable to the entire population

d. Causation, generalizable to the entire population

32 **[Intermediate]** • **Target: Blocking and Scope of Inference**

In a study on medication use and symptom severity, a simple random sample of 476 people is selected from the population of interest; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population

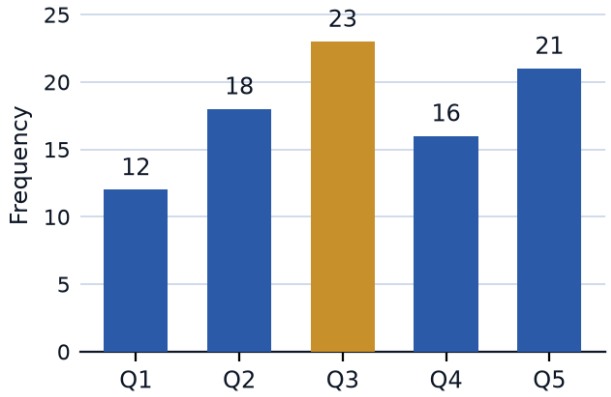
c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

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33 **[Intermediate]** • **Target: Observational studies vs. experiments**

In a study on supplement use and energy levels, a simple random sample of 406 people is selected from the population of interest; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?



a. Association only, and only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population

c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

34 **[Intermediate]** • **Target: The three principles of experimental design (RRCB)**

In a study on fertilizer use and crop yield, a simple random sample of 192 people is selected from the population of interest; researchers simply record whether each person already uses a new fertilizer or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

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35 [Intermediate] • Target: Blocking and matched pairs designs

In a study testing a new fertilizer, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.

a. Not blinded

b. Single-blind

c. Triple-blind

d. Double-blind

36 [Intermediate] • Target: Placebo effect and blinding

In a study testing a tutoring program, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.

a. Single-blind

b. Not blinded

c. Triple-blind

d. Double-blind

37 [Intermediate] • Target: Scope of inference

A study on a new exercise program first separates subjects into groups based on sex, then randomly assigns subjects to treatment or control within each group separately. Identify this experimental design.

a. Randomized block design

b. Matched pairs design

c. Completely randomized design

d. Stratified design

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Putting it together: evaluating a study design • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A study on a meditation app randomly assigns 83 subjects directly to treatment or control groups, with no grouping by any other characteristic beforehand. Identify this experimental design.

a. Randomized block design

b. Stratified design

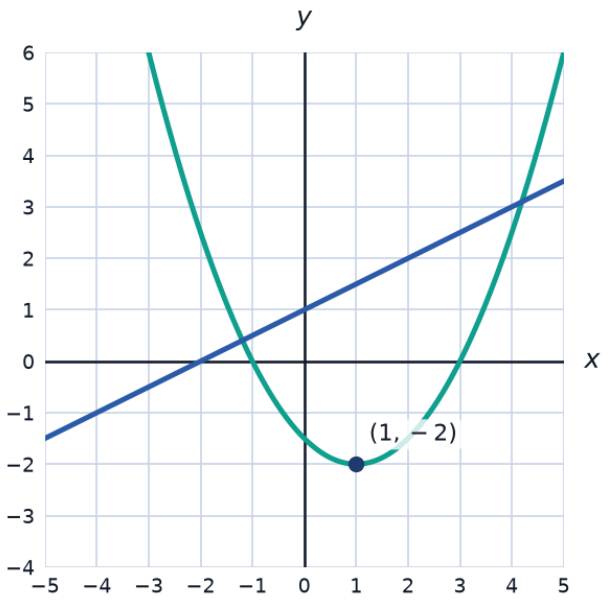
c. Completely randomized design

d. Matched pairs design

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39 [Challenging] • Target: Confounding variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A study on a new medication compares volunteers who chose to use it to volunteers who chose not to, without randomly assigning anyone to either group. Identify the main design flaw.



a. Lack of replication (too few subjects per group)

b. Lack of random assignment (confounding is likely)

c. Lack of a control group (no baseline for comparison)

d. Lack of blinding (placebo effect not controlled)

40 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 2.01 Sampling Methods & Bias

A study on a tutoring program gives every single subject the treatment, with no separate group left untreated for comparison. Identify the main design flaw.

a. Lack of a control group (no baseline for comparison)

b. Lack of random assignment (confounding is likely)

c. Lack of replication (too few subjects per group)

d. Lack of blinding (placebo effect not controlled)

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41 [Challenging] • Target: Observational Studies vs. Experiments • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

What essential feature distinguishes a randomized experiment from an observational study?

a. Treatments are actively imposed on experimental units by the researcher

b. A larger sample size is used

c. Participants are randomly surveyed by phone

d. Data is collected over a longer time period

42 [Challenging] • Target: Observational Studies vs. Experiments • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Only which type of statistical investigation allows researchers to establish cause-and-effect conclusions?

a. A well-designed randomized comparative experiment

b. A large-scale stratified observational study

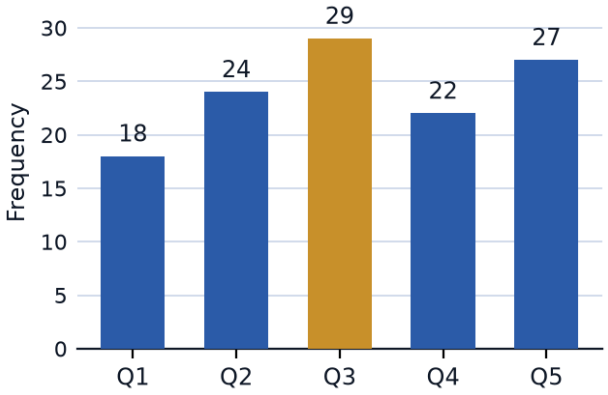
c. A voluntary response poll with $n > 50,000$

d. A census of the population

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43 [Challenging] • Target: Principles of Experimental Design • Cross-Lesson: 3.03 The Normal Distribution

In experimental design, what does 'replication' mean?



a. Assigning multiple experimental units to each treatment group

b. Repeating the entire experiment in another country

c. Using identical equipment across all trials

d. Testing subjects multiple times in a row

44 [Challenging] • Target: Blinding, Placebos, and Confounding • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

What is the 'placebo effect'?

a. When subjects show improvement simply from the psychological expectation of receiving a treatment

b. When a drug causes adverse side effects in control patients

c. When experimental units drop out of a clinical trial

d. When researchers unconsciously bias results

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45 [Challenging] • Target: Blocking and Scope of Inference • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

When should an experimenter implement a randomized block design?

a. When there is an identified characteristic expected to affect the response, so subjects are grouped by that trait before randomization

b. When the sample size is too small for simple random assignment

c. When double-blinding is completely impossible

d. To avoid having to use a control group

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